



Society of Petroleum Engineers

**SPE-229994-MS**

## **Impact of Pore-Scale Geometry on Waterflooding Efficiency: A Computational Fluid Dynamics-Based Investigation Using Direct Numerical Simulation**

Mian Umer Shafiq, School of Mining and Geosciences, Nazarbayev University, Astana, Kazakhstan; Sajjad Aziz, Department of Petroleum and Natural Gas Engineering, China University of Petroleum, Changping, Beijing, China; Muhammad Jawad Khan, Petroleum Engineering Department, Universiti Teknologi PERONAS, Perak, Malaysia; Zafar Khan, Department of Geosciences, China University of Petroleum, Changping, Beijing, China

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229994-MS](https://doi.org/10.2118/229994-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

---

### **Abstract**

This study investigates the role of pore-scale geometry (pore size, throat dimensions, connectivity) and oil-wet conditions in governing waterflooding efficiency. By analyzing three distinct pore networks (coordination numbers 3, 4, 6), the work establishes how pore architecture and wettability dictate the dynamic interplay between capillary and viscous forces, linking microscale interfacial dynamics to macroscopic displacement patterns. The scope provides actionable insights for optimizing water flood designs in heterogeneous, oil-wet reservoirs through pore-topology-aware simulation strategies. High-resolution computational fluid dynamics (CFD) simulations were conducted in the OpenFOAM solver, employing SnappyHexMesh to generate Cartesian grids for three-pore geometries with coordination numbers 3, 4, and 6. Direct Numerical Simulation resolved multiphase flow dynamics via the Volume of Fluid (VOF) method, incorporating wettability, capillary forces (modeled by the Young-Laplace equation), and viscosity contrasts. The pressure implicit with the splitting of the operator's algorithm coupled pressure-velocity fields, while a compression term in the VOF advection equation minimized interface coverage. The Brackbill number constrained time steps ( $\Delta t = 10 \mu s$ ) to suppress spurious currents. To analyze displacement efficiency, post-processing quantified saturation profiles, pressure evolution, and velocity fields. Simulations demonstrate that pore connectivity and diameter critically influence displacement efficiency. Higher coordination numbers (e.g., 6-sided connected pores) enhance sweep efficiency by enabling capillary-driven flow pathways, accelerating water propagation after initial imbibition. In oil-wet porous media, small-pore geometries resist water infiltration but sustain capillary-driven drainage, while larger pores favor viscous displacement. Pressure profiles reveal transient dominance of capillary forces over viscous forces, particularly in low-connectivity networks. The 6-connected network achieved 50% oil saturation decline and superior displacement efficiency due to optimized capillary-driven pathways compared to low-connectivity systems. These findings highlight that macroscopic recovery is governed by pore-scale architecture, necessitating geometry-aware reservoir models to predict fluid behavior. The study

concludes that optimizing waterflood strategies in oil-wet, heterogeneous formations requires prioritizing pore connectivity and alignment of capillary-viscous force balances. This work provides novel insights by coupling DNS-level CFD with diverse pore geometries to quantify how coordination number and oil-wet conditions dictate capillary-viscous force interaction. Unlike homogenized models, it establishes a predictive link between pore-scale topology and recovery efficiency, offering guidelines to tailor waterflood designs to reservoir-specific pore architecture—a critical gap in current simulation practices.

**Keywords:** Pore-connected geometry, waterflooding, computational fluid dynamics, porous media, oil recovery

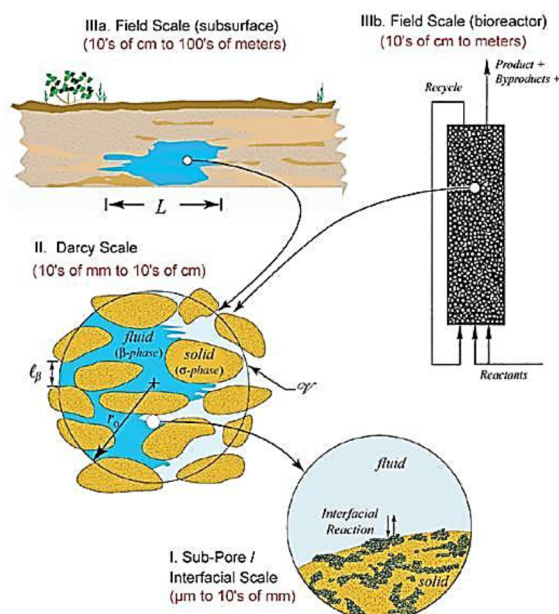
## Introduction

One of the biggest challenges faced by the petroleum industry is recovering oil from reservoirs. Energy security and resource management have been significantly impacted by improved extraction methods (Elrahmani et al., 2022). Water flooding is one of the most common secondary recovery techniques. It pushed the oil into the production wells and displaced the oil by pumping water into the reservoirs, which helps in maintaining the reservoir pressure (Qu & Liu, 2012; Ibrahim & Bowden, 2021; Khan, Aziz, et al., 2024). One of the major factors influencing the fluid flow behaviour is the pore-scale geometry, i.e., size, shape, and connectivity of pores and throats. Fluid flow behavior includes residual oil saturation, capillary pressure, and relative permeability (G. Liu et al., 2023; Tian et al., 2021). Smaller diameter pores can significantly affect Oil recovery because they can decrease the capillary pressure, which reduces the water-wetness (Khan, Mahmood, et al., 2024). Waterflooding efficiency is further affected by the irregular or complex pore structures, which change the flow paths and sweep efficiency of the reservoir (Chen et al., 2025; Qu & Liu, 2012; Su, Chai, Wang, Cao, Gu, et al., 2019).

Understanding of the multiphase flows in porous media depends on pore-scale processes. To enhance the effectiveness of oil recovery and improve waterflooding processes, it became necessary to understand the relationship between pore-scale geometry and the fluid flow in porous media (Ibrahim & Bowden, 2021). Changes in the pore geometry can affect different parameters in porous media, like fluid distribution, saturation levels, and capillary forces. These are the parameters that affect the efficiency of water flooding (Molins, 2015). Although simulation and modelling techniques have improved with time, the complexities of fluid flow present in these small pores continue to be a significant barrier for understanding and enhancing water flooding processes (Fagbemi et al., 2018).

Fig. 1 depicts the hierarchical scales in microbially mediated porous media processes: (I) the sub-pore interfacial scale ( $\mu\text{m}$ - $\text{mm}$ ), where reactions take place; (II) the Darcy scale ( $\text{mm}$ - $\text{cm}$ ), which illustrates the distribution of fluid-solid within pore networks; and (III) field scale applications in subsurface environments and bioreactors ( $\text{cm}$ - $\text{m}$ ). This framework links macroscale transport processes to microscale biological activities (Molins, 2015).

Usually, core-scale experiments and simulations skip small details of fluid dynamics at the pore scale level (Szűcs et al., 2023). However, recent developments in computational power and numerical techniques have made DNS and CFD effective tools for exploring pore-scale phenomena (Frank et al., 2018; Su et al., 2019). These approaches accurately compute fluid flow at the pore scale, accounting for the effects of wettability alteration, pore geometry, and interfacial tension on multiphase flow behavior (Farhadzadeh et al., 2024; Yang et al., 2016). Pore-scale models provide the necessary resolution to study fluid-rock interactions in porous media (Mehmani & Tchelepi, 2017).



**Figure 1—The conceptual progression of length scales related to (microbially driven) interfacial reactions in porous media (Molins, 2015).**

The effect of pore shape on the performance of water flooding is investigated using CFD simulations. This study conducts pore-scale analysis to examine how oil-water displacement is influenced by pore geometry, including the coordination number (the number of neighboring connected pores) and other geometric characteristics. Complex porous media are numerically discretized using SnappyHexMesh, a meshing utility available in the open-source CFD toolbox OpenFOAM. Fluid flow through various pore geometries is modeled using the VOF method combined with DNS.

The primary objective is to develop a mathematical simulation model that accurately represents flow displacement at the pore scale in heterogeneous porous media. To achieve this, image processing techniques, including segmentation, are used to determine pore size distribution from 2D images. These geometries are then integrated into CFD simulations to study fluid displacement mechanisms in complex structures.

The study examines three distinct pore geometries with varying coordination numbers, evaluating the effects of capillary forces, wettability behavior, and pore connectivity on water flooding efficiency. The findings enhance understanding of the relationship between pore-scale characteristics and flow performance, potentially leading to improved oil recovery strategies. Ultimately, this research aims to advance fluid flow modeling in porous media, supporting more efficient reservoir management and hydrocarbon recovery.

## Literature Review

The pore-scale seepage mechanism was investigated using experiments and lattice Boltzmann simulations to evaluate the influence of wettability on fluid flow and pore geometry (Zhang et al., 2022). Another study performed waterflooding experiments on fractured and non-fractured micromodels, identifying optimal injection rates determining optimal injection rates of 0.15 mL/min for fractured models and 0.1 mL/min for non-fractured models. Their findings indicate that fractures no longer increase fingering and reduce recovery through early water breakthrough. Circular models achieved higher oil recovery via compact displacement, while square oil-wet models showed lower recovery due to capillary resistance and oil films, shown in Fig. 2 (Tian et al., 2021).

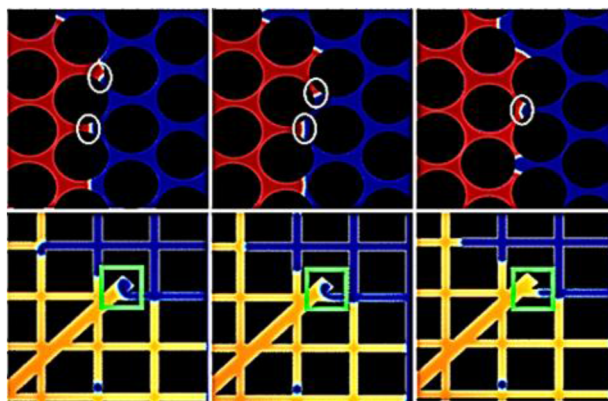


Figure 2—Wettability profile of the reservoir.

A quasi-static pore-scale modeling technique is used in this work to analyze the behavior of two-phase fluids during drainage and imbibition at low capillary number equilibria. A new object-based up-scaling technique is used to apply the model to 3D pore structures with different contact angle distributions, both consolidated and unconsolidated. The simulations provide predictions for electrical and hydraulic conductivity, as well as capillary pressure and relative permeability curves. Results emphasize the critical role of wettability and reveal how rock configuration, cementation, and phase saturation influence fluid flow and electrical pathways within geological formations (Mohammadmoradi & Kantzas, 2016).

Another common example of this phenomenon is wettability heterogeneity, which arises from variations in initial water saturation due to capillary rise above the oil-water contact or differences in the properties of reservoir rock types (Fig. 3) (Jackson et al., 2003).

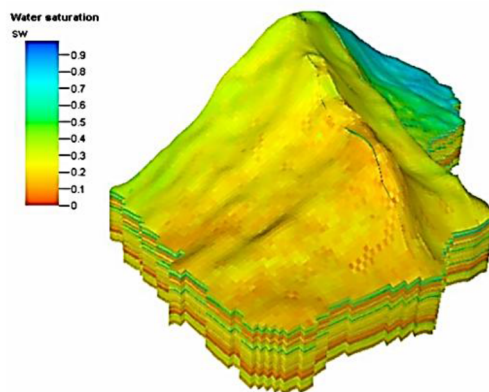


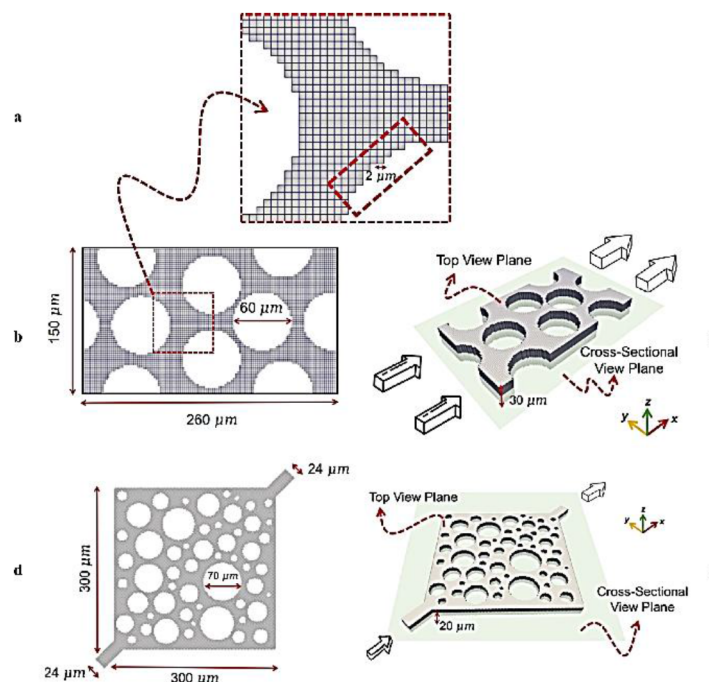
Figure 3—A field-scale demonstration shows how lateral and vertical variations in initial water saturation can influence the saturation function (Jackson et al., 2003).

Low-salinity water injection has also shown incredible results as a potential oil recovery method, showing promising results compared to conventional high-salinity (HS) waterflooding (Khan et al., 2021). However, conflicting results from both laboratory and field studies have hindered a clear understanding of the underlying mechanisms governing the crude oil-rock-brine interfaces during LS water injection (Khan et al., 2018). Mahmud et al (2007) modelled how low salinity (LS) brine pushes oil through porous media. This research shows that reducing salt level changes how water wets a rock's surface, which affects the trapped oil and how it can reduce residual oil saturation in a reservoir, while the LS effectiveness depends on factors like initial wettability, pore size distribution, and end-point capillary pressure (Mahmud et al., 2007).

In mixed-wet reservoirs, corner flow significantly impacts low-salinity waterflooding (LSWF) efficiency by transporting low-salinity water ahead of the fluid front, shifting displacement from piston-like to multi-



directional flow. OpenFOAM simulations reveal a 22% oil recovery difference between 2D and quasi-3D models, and this phenomenon is clearly shown in Fig. 4. The interaction between solute transport and wettability alteration creates new oil trapping mechanisms, reducing LSWF efficiency in mixed-wet systems despite improved recovery over high-salinity flooding (Ahmadi-Falavarjani et al., 2024).



**Figure 4—Low-salinity waterflooding in mixed-wet systems as simulated at the pore scale: effects of corner flow, surface imperfections, and wettability change kinetics (Ahmadi-Falavarjani et al., 2024).**

Nonuniform mixing in LSWF depends on pore geometry and flow patterns. Wettability alteration (WA) alters capillary pressure, which can mobilize trapped oil. Both experimental (pore-doublets) and numerical simulation methods examined how wettability alteration, salinity distribution, and initial wettability affect oil recovery. In oil-wet systems, LSWF can remobilize trapped oil, with the salinity threshold model showing the highest recovery. The interplay of capillary forces, viscosity, and dispersion influences whether oil is recovered or re-trapped (Namaee-Ghasemi et al., 2021).

A CFD-based approach consisting of four types of Micro-CT images is being used to generate structured mesh models for two-phase immiscible displacement. The focus is on the critical capillary number at micro- and macro-scales and its impact on relative permeability, oil recovery, and displacement efficiency. Results shown in Fig. 5 clearly show the residual oil saturation at different capillary numbers in the range of  $10^{-6}$  to  $10^{-5}$  at the micro scale (Wang et al., 2019).

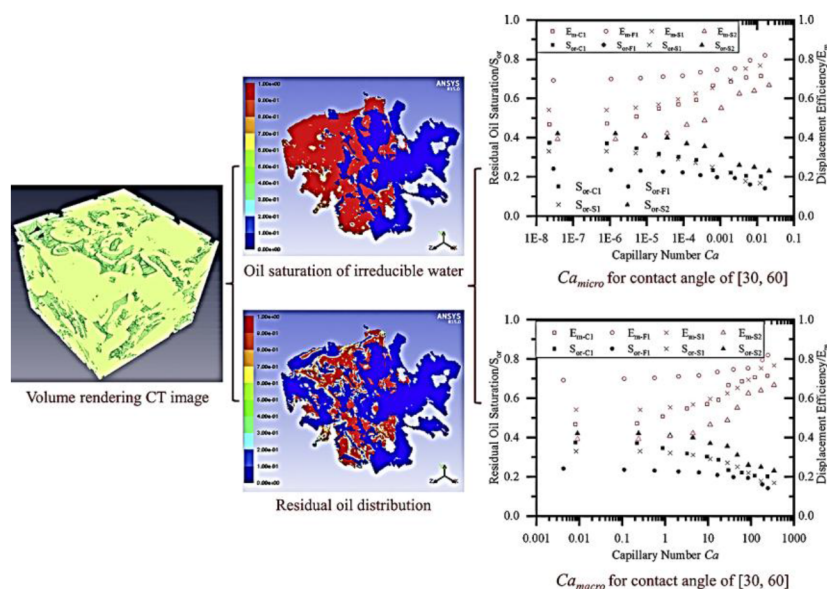


Figure 5—CT scan image of residual oil saturation and capillary numbers (Wang et al., 2019).

This study employed the flow experiments and X-ray micro-tomography to examine wettability alteration and fluid distribution during LSWF in carbonate rock. Brine injection resulted in approximately 85% oil efficiency, attributed to a wettability shift from oil-wet to intermediate-wet conditions. A reduction in brine salinity enhanced oil recovery by modifying fluid-rock interactions (Salem et al., 2021). As shown in Fig. 6, decreased salinity alters wettability relative to baseline water, leading to improved oil displacement. The LSWF study was conducted using a calcite micromodel, and the results indicate that lowering brine salinity can improve oil recovery through wettability alteration, with connate water potentially amplifying this effect (Mohammadi & Mahani, 2020).

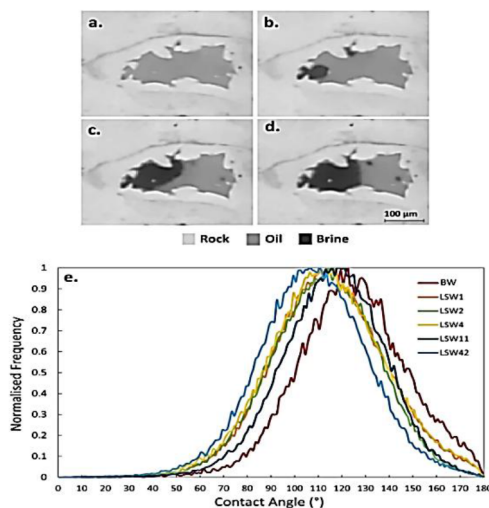


Figure 6—Fluid-rock interactions in an oil-wet system before (a) and after low-salinity water injection (b–d). Histogram (e) shows contact angle distribution measured from segmented 3-D X-ray micro-tomography images (Salem et al., 2021).

This study uses high-resolution synchrotron X-ray tomography to examine pore-scale processes in multiphase porous media made of oil-wet glass beads. The system consisted of oil-wet glass beads, which were imaged to get the pore body/throat structure. Among the significant results is that large pore spaces with high aspect ratios and coordination numbers mostly trap nonwetting phases. These detailed pore-scale

insights can improve multiphase flow models, helping to better understand fluid displacement and enhance predictions in porous media systems (Al-Raoush & Willson, 2005).

Fagbemi et al. focus on the interaction between fluid flow and porous media with complex geometries at the pore scale. It uses a finite element method (FEM) to model the deformation of rock material under stress from fluid forces in 3D porous media, including Berea sandstone and polydisperse sphere packs. The study explores how boundary stresses influence permeability and deformation in both real and synthetic porous media, particularly at low and moderate Reynolds numbers (Fagbemi et al., 2018).

The above Fig. 7 shows a 3D visualization of porous media structures and their computational representations. (a) A 2D binary slice showing the irregular pore-grain interface pattern; (b-d) 3D volume reconstructions of a porous medium sample with increasing levels of processing, transitioning from raw segmented data to meshed domains; (e-g) Different representations of spherical packing models highlighting flow paths through the pore spaces; and (h-j) Progressive tetrahedral mesh refinements used for computational fluid dynamics or geomechanical simulations of the porous structure. These visualizations demonstrate the workflow from tomographic imaging to computational model generation for analyzing transport and mechanical properties in heterogeneous media. It is exhibited that particle shape and pore geometry significantly affect deformation and permeability changes, with Berea sandstone experiencing more deformation than sphere packs due to surface roughness and pore structure (Fagbemi et al., 2018). Pore geometry is affected by the permeability in a porous medium, as Liu & Jeng investigate the impact of particle geometry on the permeability of porous media. Using the spherical harmonic function and Cellular Automata model, particles with varying shapes and surface features were constructed. The Lattice Boltzmann Method (LBM) simulated fluid flow, showing that geometry significantly affects permeability (Y. F. Liu & Jeng, 2019).

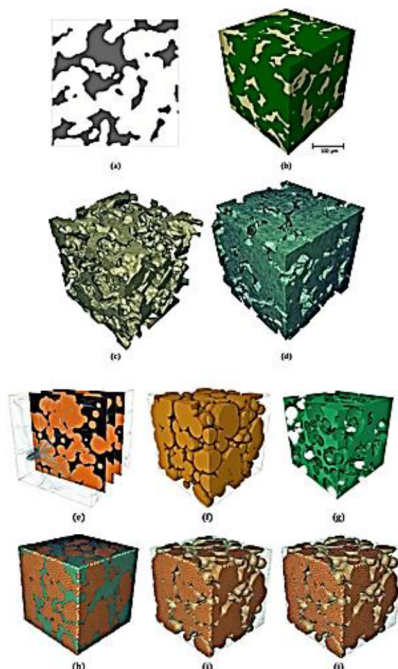


Figure 7—(a) A  $350 \times 350 \times 350$  voxel grid, (b) a 2D slice of the image, (c) a 3D representation of the solid grains, and (d) the tetrahedral mesh structure for both pore and matrix spaces. (e) An image of a sphere, (f) raw 2D image slices showing flow streamlines, and (g) a 3D image with intersecting spheres to maintain stress continuity in the pore space geometry. The tetrahedral meshes for the pore and matrix spaces are displayed in Figures 1h to 1j (Fagbemi et al., 2018).

## Methodology

With OpenFOAM, images are changed into computational meshes using the SnappyHexMesh tool. This raises the mesh to the porous media 1, 2, and 3, which show complex pore structures at the molecular scale in Figs. 8, 9, and 10. The pore geometry with six pore throats is shown as porous media 1 in Fig. 8, four pore throats make up porous media 2 in Fig. 9, and three pore throats represent porous media 3, the final pore geometry in Fig. 10.

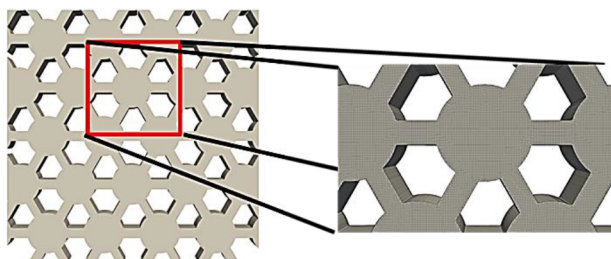


Figure 8—Porous media 1 - Coordination no. of 6.

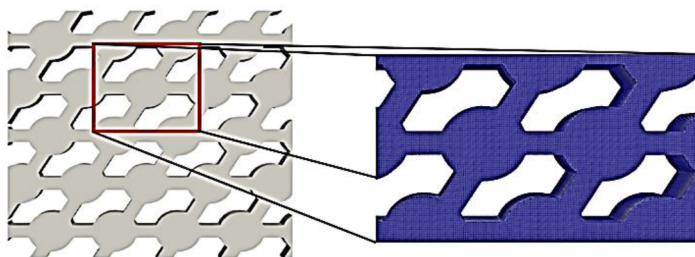


Figure 9—Porous media 2 - Coordination no. of 4.

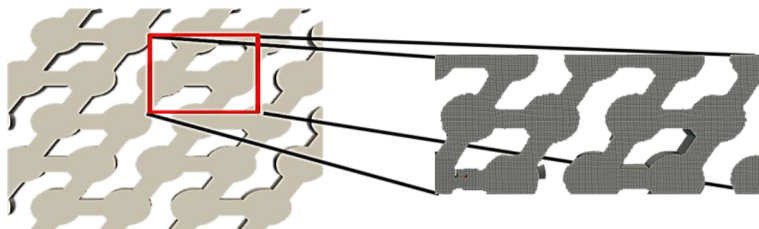


Figure 10—Porous media 3 - Coordination no. 3.

After that, numerical solvers that incorporate capillary effects are used to create pore-scale flow simulations. As a result, the interactions between the injectant and displaced fluids can be better understood. To comprehend the operation of pore-scale displacement mechanisms and the critical elements in fluid displacement, such as pore size distribution, the simulation results are closely analyzed. And lastly, the interpretations aim to provide methods for enhancing microscopic sweep efficiency as well as techniques to maximize oil recovery through the waterflooding approach.

## Mathematical Model

A fluid dynamics model is key for simulating multiphase flow at the pore scale. This includes applying governing equations like the Navier-Stokes equations for fluid flow, setting boundary conditions, and factoring in interactions between fluids and solids (Hoang et al., 2013; Su, Chai, Wang, Cao, Yu, et al., 2019). DNS of multiphase flow provides the most accurate predictions of pore-scale fluid dynamics, though it comes with a high computational cost (Mehmani & Tchelepi, 2017). DNS is a precise method where all scales are completely resolved, requiring a very fine grid (Li & Pistorius, 2023; Wood et al., 2015). In DNS,



the Navier-Stokes equations are solved separately for each fluid phase, with the right boundary conditions set at fluid-fluid interfaces to maintain continuity in the velocity field. These circumstances govern the dynamics of isothermal flow in immiscible and incompressible multiphase systems. The Young-Laplace equation is used to determine interfacial stress gradients.

By capturing interface dynamics explicitly, this method becomes computationally challenging because of the complexity of solving a moving boundary problem. To tackle this, the VOF method treats the two fluid phases as a single effective mixture. This is done by introducing a scalar color function,  $\alpha \in [0,1]$ , which represents the volume fraction of one phase within a computational cell. An  $\alpha$  value of 0 or 1 means there is a pure single phase, while values between 0 and 1 indicate the presence of an interface between the two immiscible fluids. The VOF method then uses the following equations to ensure that mass and momentum are conserved:

$$\nabla \cdot \mathbf{U} = 0 \quad (1)$$

And

$$\frac{\partial(\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U} \mathbf{U}) = -\nabla p + \nabla \cdot \mu (\nabla \mathbf{U} + \nabla \mathbf{U}^T) + F_b + F_c \quad (2)$$

Where  $F_b$  and  $F_c$  represent the external body (such as gravity) and capillary forces, respectively, while  $\mathbf{U}$ ,  $t$ , and  $p$  represent velocity, time, and pressure. Superscript  $T$  stands for transposition. When analyzing a single mixture, its characteristics are taken to be a linear combination of the two fluids that make it comprised; for instance, the mixture's density ( $\rho$ ) and dynamic viscosity ( $\mu$ ) are indicated as the wetting phase by  $\alpha = 1$ .

$$\rho = \rho_w \alpha + \rho_n (1 - \alpha) \quad (3)$$

$$\mu = \mu_w \alpha + \mu_n (1 - \alpha) \quad (4)$$

While capillary forces are calculated as follows

$$F_c = \sigma k n_I \delta_I \quad (5)$$

The interface curvature is calculated as

$$k = -\nabla \cdot n_I = -\nabla \cdot \frac{\nabla \alpha}{|\nabla \alpha|} \quad (6)$$

And for the color function, the advection equation is used as follows.

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\mathbf{U} \alpha) = 0 \quad (7)$$

where  $\delta_I$  is a Dirac  $\delta$  function that is used to limit the capillary forces to acting just at the interface,  $\sigma$  is the surface tension,  $k$  is the interface curvature, and  $n_I$  is the unit normal to the interface.

## Numerical Implementation

Similar to several other numerical techniques, VOF experiences numerical diffusion and interface smearing, which result from resolving the color function's discretized advection equation. To mitigate this numerical artefact, Eq. (7) is modified by including an extra component  $\alpha (1 - \alpha) U_r$ .  $U_r = \min [\alpha U_{cv}, \max (\mathbf{U}d)]$  is an approximation of the so-called compression velocity (van Leer, 1979). Based on the research of (Hoang et al., 2013; Deshpande et al., 2012),  $\alpha$  is the compression coefficient set to one. The velocity in a particular control volume is denoted by  $U_{cv}$ , whereas the maximum velocity throughout the porous medium is indicated by  $\max (\mathbf{U}d)$ . The compression velocity is limited to acting only at the interfaces by the phrase  $\alpha (1 - \alpha)$ .

The pressure implicit with the splitting of operators (PISO) approach is used to solve the pressure-velocity coupling in the Navier-Stokes equations. We initially apply a rectangular mesh with a cell size of  $x = y = 13.5 \mu\text{m}$  to construct the discretized pore space for the simulations. We then remeshed using OpenFOAM's

SnappyHexMesh module. The Brackbill number  $tBk = (\rho_{avg} \Delta x^3 / \pi \sigma)^{1/2}$ , where  $x = 13.5 \mu m$  is the cell size and  $\rho_{avg}$  is the average fluid density in the domain, is used to determine the time step size. We set the time step size to  $t \approx 3 tBk$ , which for our parameters yields  $t = 10 \mu s$ , to decrease the computing runtime.

Additionally, the formulation is costly to compute, and the available open-source solvers work only with flows that have Cartesian meshes. Other common VOF formulations include the sharp surface force (SSF), which sharpens the interfacial area and limits where interfacial forces act, and the filtered surface force (FSF), which gathers and filters the capillary fluxes (Raeini et al., 2012). FSF and SSF consider several heuristic characteristics that may change depending on the situation (Pavuluri et al., 2018). This study reduces the effect of false currents by using a fine mesh along with short time steps based on  $tBk$ . False velocities do not consistently focus on specific control volumes when solving Eq. (7).

### Discretizing the Porous Medium Model

Discretization is a crucial stage in the solution of porous media models. It transforms the controlling partial differential equations into a system of computer-solvable algebraic equations. The study procedure was set up to guarantee pore-scale modelling consistency. The samples' unique pore structure was first reflected in a computational grid created using OpenFOAM's SnappyHexMesh module. To provide reliable computer models for in-depth investigations of fluid movement in porous media, the mesh sizes were modified, and the cells were optimized. The OpenFOAM framework's SnappyHexMesh tool is frequently utilized for this purpose (Renard et al., 2024). SnappyHexMesh is an automated meshing tool that produces hexahedral dominant meshes, which effectively resolve the complex pore structures of porous media (Kyrloglou et al., 2024).

### Simulating Displacement

Transport and fluid movement in the model are analysed by numerical simulations. At first, the boundary and initial conditions were modelled as part of the displacement simulation. Pressure gradients, flow rates, and fluid characteristics have been used in this step. Selecting the appropriate numerical solver, like interFoam, in the OpenFOAM environment is an essential step in choosing the best CFD technique. This decision was based on the flow regime and fluid characteristics. Multiphase flow processes, fluid-fluid interactions, and interfacial behaviour were all accurately represented by the models. The variations in flow inside the porous media make these procedures more difficult.

In analyzing the results, fluid flow patterns, saturation distributions, and displacement dynamics using OpenFOAM's post-processing tools were visualized. Pore-scale simulation modeling requires a consistent research process that considers the complex pore structure of samples. The SnappyHexMesh library in OpenFOAM sets up a computational grid for thorough investigations of fluid movement in porous media. The relevant parameters, including fluid saturations, velocity, and pressure distributions, were quantified to improve the understanding of the simulated processes. What fluid configurations and displacement fronts change over time was also explored, which provided insights into the dynamic behavior of fluid movement within the porous media.

## Results and Discussions

This study analyzes water flooding dynamics in oil-wet reservoirs at the pore scale, focusing on the impact of pore connectivity. Three pore network geometries, including coordination number 3, coordination number 4, and coordination number 6, are analyzed to assess how capillary forces, pore connectivity, and wettability influence multiphase flow. Numerical simulations, conducted using computational fluid dynamics, track temporal changes in water saturation, pressure distributions, oil displacement, and velocity profiles. The results highlight the significant role of pore-scale characteristics in governing macroscopic oil recovery efficiency.

### Water saturation dynamics in oil-wet systems with varying geometries

Fig. 4.1 illustrates the temporal evolution of water saturation within oil-wet porous media 1, 2, and 3 at different time snapshots. In porous media 1 and 2 (Figs. 4.1a-f), water invasion begins at the left boundary (inlet) and gradually propagates through the pore network. At the initial time step (Figs. 4.1a and 4.1d,  $t_0 = 0.01$  s), water saturation is limited to a narrow inlet region (green), while the rest of the domain remains predominantly oil-saturated (red). This is expected in oil-wet systems, where the affinity of pore walls to oil increases capillary entry pressure for water, thereby resisting displacement.

By intermediate time steps ( $t_1$ ), shown in Fig. 11b and 11e, water begins to encroach further into the matrix, although progress remains localized due to capillary resistance and unfavorable wettability conditions. At  $t_2$  (Fig. 11c and 11f), water saturation extends deeper into the porous domain, but large oil-filled regions persist. The slower advancement in porous media 2 compared to media 1 could be attributed to reduced pore connectivity and more tortuous flow paths.

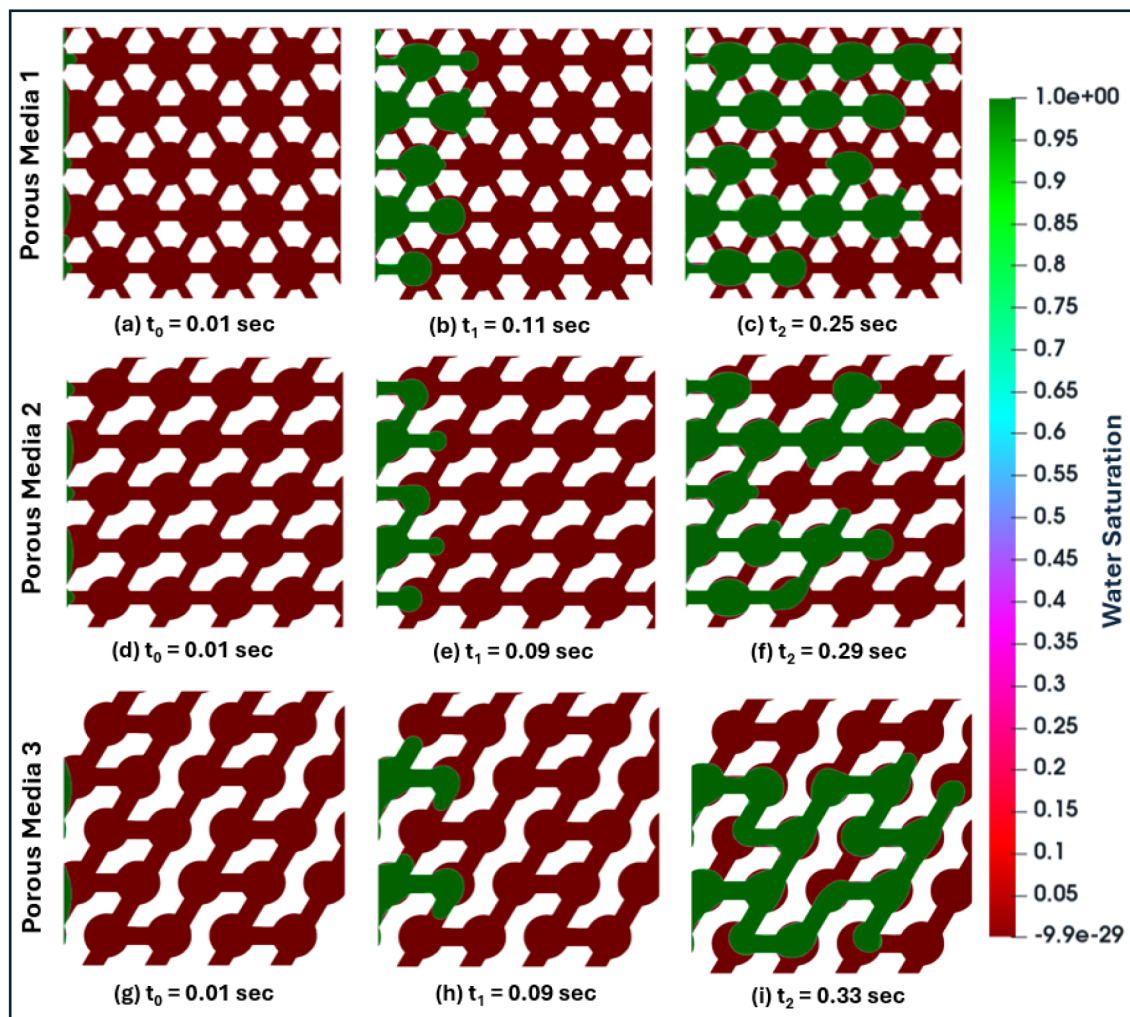


Figure 11—Water saturation profiles in oil-wet porous media with varying geometries: porous media 1 (hexagonal structure, coordination number 6), porous media 2 (tetragonal structure, coordination number 4), and porous media 3 (triangular structure, coordination number 3).

Porous media 3 (Fig. 11 g–i) exhibit a more pronounced change in saturation profile over time. Initially (Fig. 11g,  $t_0 = 0.01$  s), water entry is confined to the inlet. By  $t_1$  (Fig. 4.1h), water has advanced significantly, indicating a relatively lower capillary entry barrier, potentially due to better connectivity or more favorable

pore throat dimensions. At  $t_2$  (Fig. 11i), a near-complete water invasion is observed, although some isolated oil pockets remain, highlighting the persistent influence of capillarity in oil-wet media.

Overall, these saturation profiles indicate the combined effects of oil-wet wettability, capillary-driven resistance, and pore network geometry on water flooding efficiency. The progression from initial invasion to deeper penetration is non-uniform and reveals how heterogeneities in structure can critically delay or limit water breakthrough, even under constant injection conditions.

### Oil saturation trend in pore connectivity effects

A quantitative evaluation of oil saturation over time enables a direct comparison of displacement efficiency and recovery dynamics in each structure. Fig. 12 presents the variation of oil saturation ( $S_o$ ) with time for porous media 1, 2, and 3. Initially, all three systems exhibit similar oil saturation values around 0.95. As water invades the pore space, a gradual reduction in oil saturation is observed across all cases, though the rate and extent of reduction differ significantly due to geometric variations and coordination number.

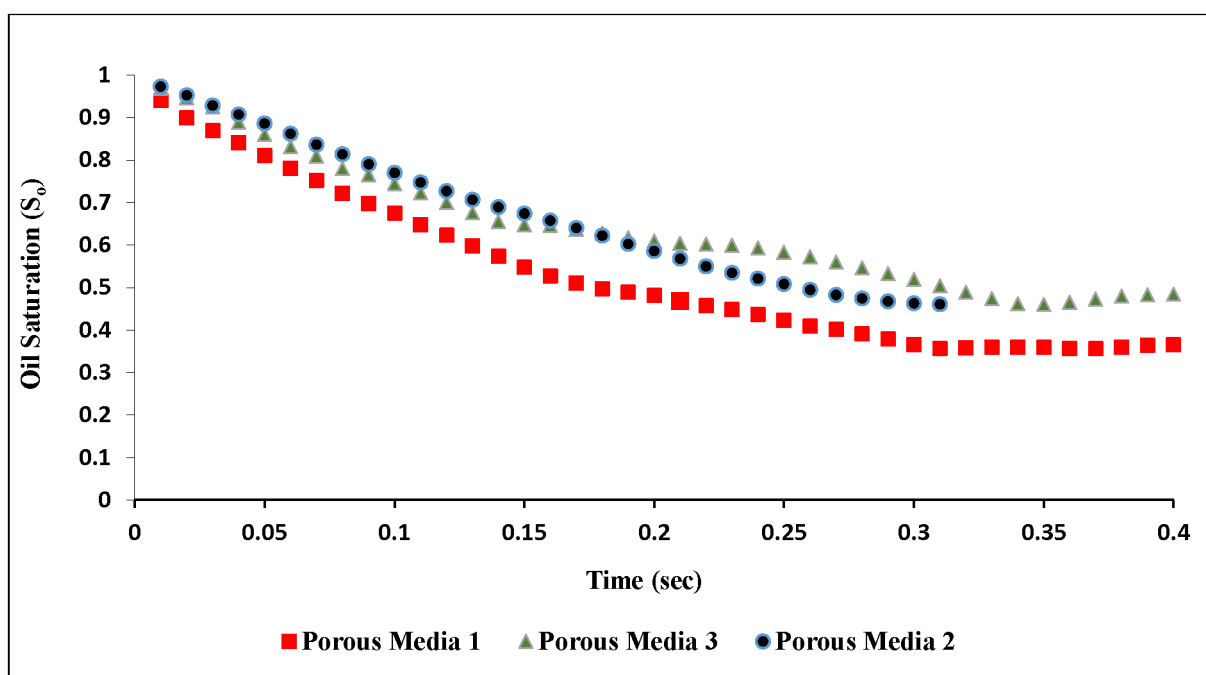


Figure 12—Oil saturation as a function of time for oil-wet systems with varying pore geometries.

Porous media 1, with the highest coordination number (6), shows the most efficient oil displacement. Its oil saturation declines steadily throughout the simulation, reaching approximately 37% at 0.4 seconds. This reduction indicates enhanced connectivity and reduced capillary resistance, facilitating continuous water invasion. In contrast, porous media 3 (coordination number 3) exhibits a more irregular trend. After an initial drop in oil saturation to around 60% by 0.2 seconds, a temporary increase is observed, likely due to local trapping of water or redirection of flow paths. Eventually, the  $S_o$  stabilizes near 50%, indicating less efficient displacement compared to porous media 1.

Porous media 2 demonstrates intermediate behavior, with smoother saturation decline and a final oil saturation value between those of media 1 and 3. The overall trend confirms that higher coordination numbers promote more uniform displacement and lower residual oil saturation. Thus, pore connectivity and geometry play a decisive role in determining the efficiency of oil recovery in oil-wet media.



### Field velocity during water invasion in oil-wet connectivity systems

The evolution of fluid velocity during the early stages of water invasion in oil-wet systems with different pore geometries is demonstrated in Fig. 13. The velocity magnitude is color-coded, with blue indicating low flow and yellow to red representing regions of higher velocity.

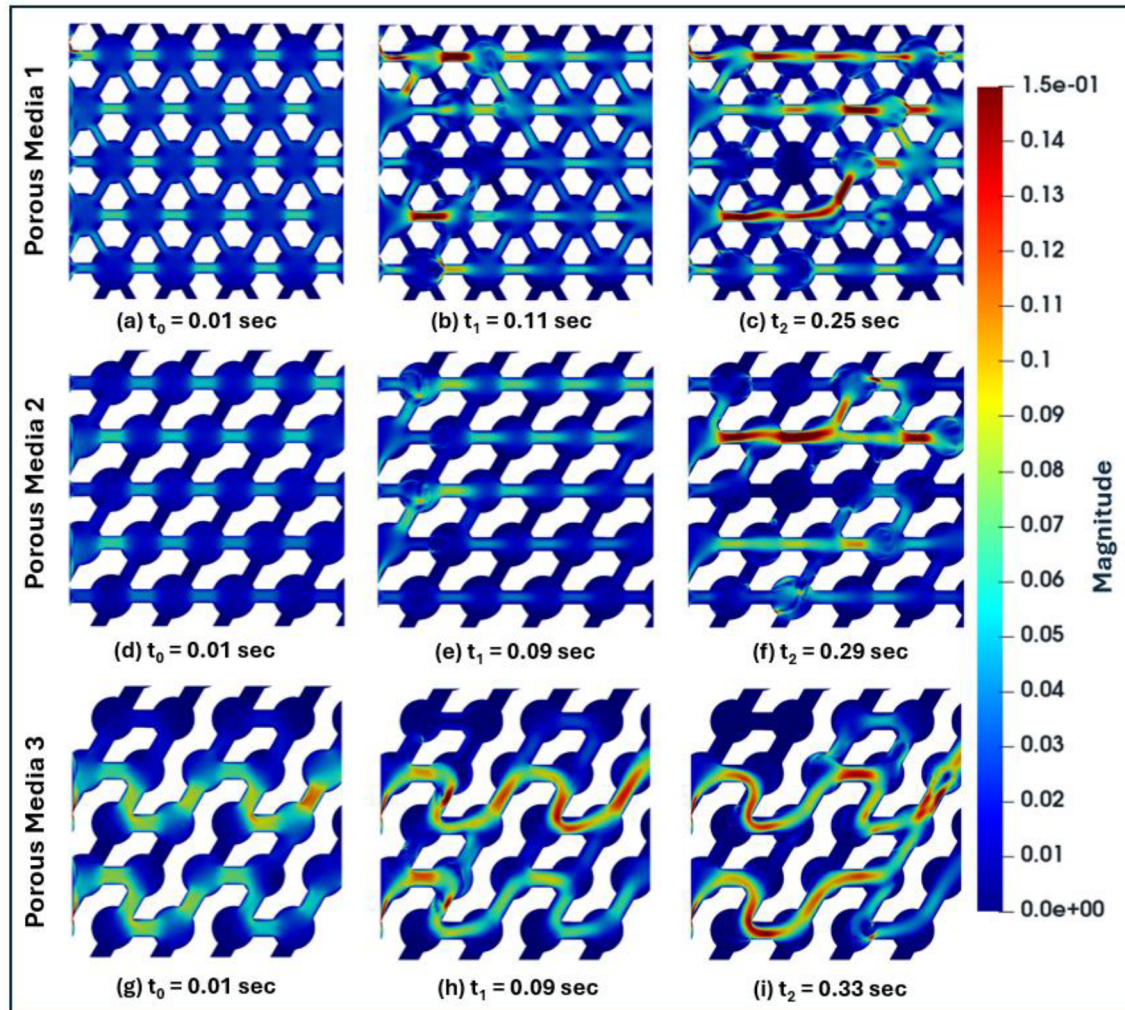


Figure 13—Velocity magnitude distribution during water invasion in oil-wet porous media 1, 2, and 3 at different time intervals. Higher coordination numbers (e.g., media 1) promote more distributed flow, while lower-coordination geometries (e.g., media 3) lead to irregular flow paths and bypassed regions due to wettability and capillary effects.

In porous media 1 (Fig. 13a–c), the velocity profile begins with very low values (predominantly blue) at  $t_0 = 0.01$  sec, reflecting the dominance of oil saturation and minimal initial water development. By  $t_1 = 0.11$  sec, moderate velocity zones (up to 0.14 m/s) emerge near the inlet, marking the onset of drainage. As time progresses ( $t_2 = 0.25$  sec), water continues to invade selectively along preferential pathways, but large portions of the medium remain static, showing the impact of oil-wet conditions and coordination number on restricting flow.

Porous media 2 (Fig. 13d–f) displays similar behavior with slightly reduced velocity magnitudes due to its lower coordination number. At  $t_2 = 0.29$  sec, localized flow channels begin to form, but the displacement front remains confined near the inlet, indicating higher resistance to fluid movement due to geometrical constrictions and capillary barriers.

In porous media 3 (Fig. 4.3g–i), the velocity profile reveals more dynamic behavior. Water invades earlier and more irregularly, forming multiple active flow paths. By  $t_2 = 0.33$  sec, high-velocity zones ( $\geq 0.12$  m/

s) extend across the domain, although some low-velocity regions persist. This uneven propagation reflects the complex interplay between pore structure, viscous forces, and capillary pressure. The emergence of preferential flow channels and bypassed zones highlights limitations in displacement efficiency despite initial faster invasion.

Overall, the velocity fields emphasize how pore geometry, wettability, and coordination number control fluid flow dynamics in oil-wet zones. High-coordination structures (e.g., media 1) facilitate broader distribution of flow paths, while lower-coordination designs (media 3) show more focused and uneven displacement fronts.

### Pressure distribution and temporal response in oil-wet porous media

The pressure distribution in porous media 1, 2, and 3 during early water flooding in oil-wet systems is shown in Fig. 14. The color scale indicates pressure magnitude, ranging from  $1.0 \times 10^5$  Pascal (dark blue) to  $1.0016 \times 10^5$  Pascal (dark red).

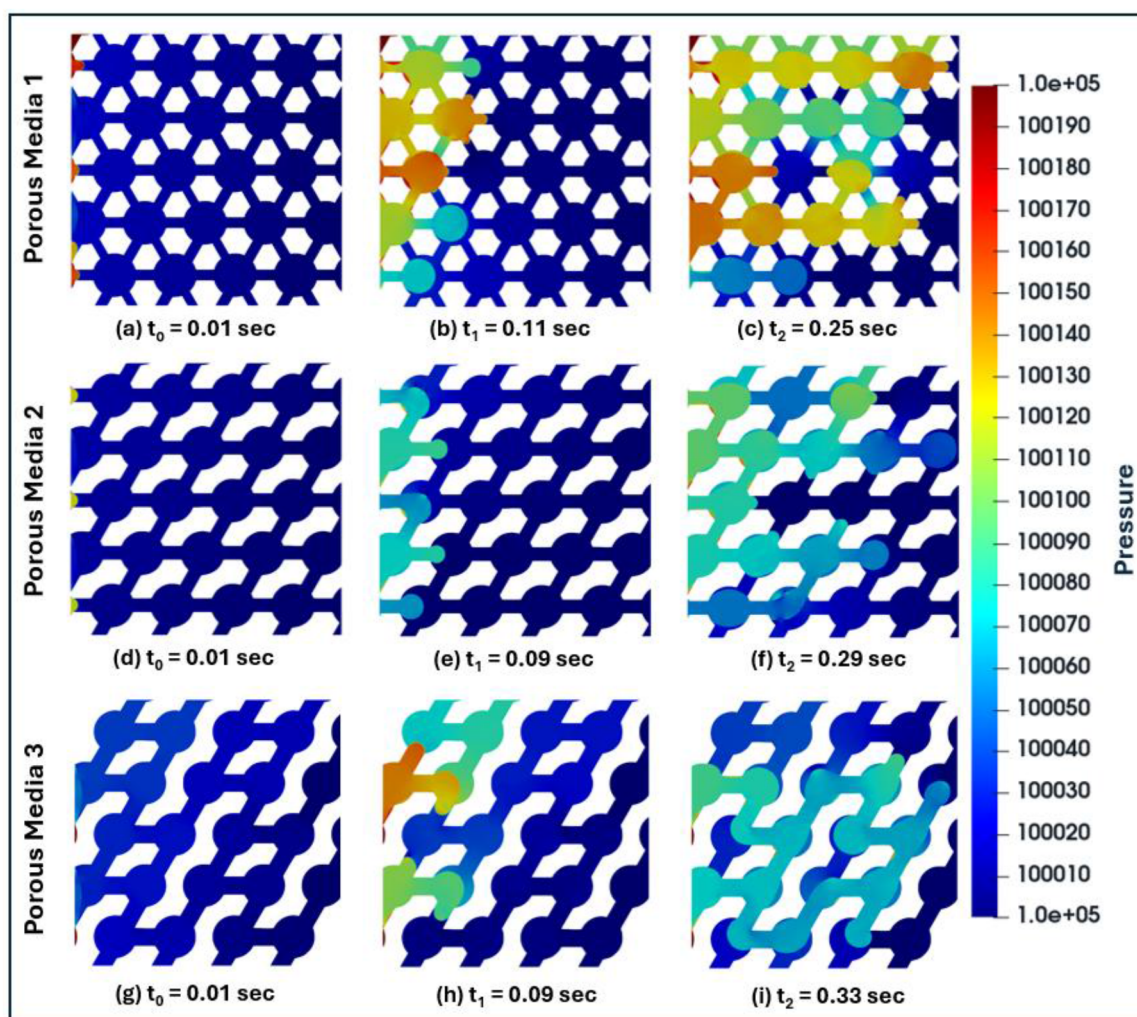


Figure 14—Pressure field evolution in oil-wet porous media 1, 2, and 3 at different time intervals. Pressure buildup near the inlet indicates resistance to displacement, while broader pressure distribution reflects advancing water fronts. Higher coordination numbers result in faster and more uniform pressure propagation.

In porous media 1 (Fig. 14a–c), the initial pressure field at  $t_0 = 0.01$  sec is nearly uniform, with slight increases near the inlet ( $\sim 100,030$  pascal), indicating the onset of water injection. As time progresses ( $t_1 = 0.11$  sec), localized pressure buildup occurs in regions where water begins displacing oil. By  $t_2 = 0.25$  sec,

the pressure front has propagated across most of the domain, with values reaching up to  $\sim 100,160$  pascal. This gradient reflects the capillary resistance encountered as water advances through oil-wet pores with high coordination.

In porous media 2 (Fig. 14d–f), the pressure distribution remains concentrated near the inlet at early times, with values ranging from 100,010 to 100,020 pascal. By  $t_1 = 0.09$  sec, the pressure begins to increase within the interior as water penetrates deeper. At  $t_2 = 0.29$  s, more uniform pressure zones develop, indicating moderate water advancement and a reduced pressure gradient compared to media 1. Porous media 3 (Fig. 14g–i) shows delayed pressure propagation, with localized buildup near the inlet at  $t_1 = 0.09$  s ( $\sim 100,060$  pascal). At  $t_2 = 0.33$  sec, a broader pressure field is established, although some low-pressure regions remain, suggesting uneven displacement and the presence of capillary barriers.

In addition to the spatial maps, Fig. 15 presents the temporal evolution of pressure in all three media. Each case exhibits damped oscillations, reflecting the interplay of viscous and capillary forces at the pore scale. Porous media 1 shows the highest-pressure fluctuations, with multiple sharp peaks during the early phase, attributed to rapid changes in flow paths and abrupt pore filling. The high coordination number allows for more dynamic fluid rearrangement, producing stronger pressure responses. Porous media 2 exhibits a smoother trend, with an initial spike followed by a gradual decline and moderate oscillations. This reflects more controlled and stable fluid displacement due to limited connectivity and fewer flow paths. Porous media 3 demonstrates a delayed pressure rise followed by a sharp peak and eventual damping. The late pressure buildup corresponds to the slow initial invasion and localized capillary barriers. Once these are overcome, the system experiences a temporary surge in pressure before stabilizing.

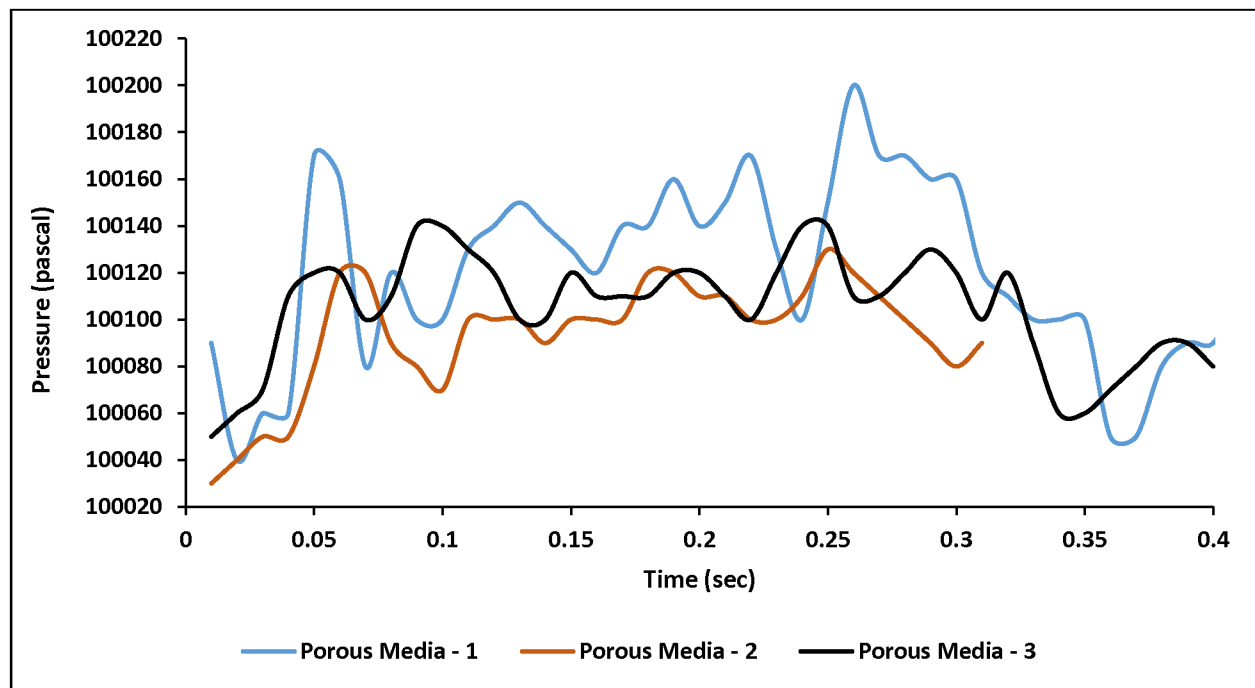


Figure 15—Pressure variation during water injection in porous media 1, 2, and 3. Damped oscillations reflect capillary-driven displacement dynamics and differences in pore connectivity.

Together, the spatial and temporal pressure analyses confirm that pore geometry and coordination number strongly influence pressure propagation and displacement efficiency. More connected networks (e.g., porous media 1) facilitate pressure equilibration and more effective oil displacement, while lower-connectivity systems (e.g., porous media 3) experience delayed and uneven pressure development.



## Conclusion

This study demonstrates that pore geometry and coordination number play a significant role in governing the efficiency of water flooding in oil-wet porous media. Through pore-scale simulations of three distinct geometries (trigonal, tetragonal, and hexagonal), we evaluated how connectivity, wettability, and capillary forces influence multiphase flow dynamics. The pressure profiles reveal that capillary forces strongly resist water penetration, particularly in low-coordination structures, resulting in delayed and uneven displacement. Conversely, higher coordination numbers exhibited extensive pressure propagation and more efficient oil displacement. The velocity and saturation profiles further confirm that pore geometry controls flow pathways and displacement front uniformity, with porous media 1 (coordination number 6) achieving faster and more complete oil displacement compared to porous media 3 (coordination number 3). These simulation results highlight the importance of incorporating pore-level structural parameters (such as coordination number, pore size distribution, and wettability) into enhanced oil recovery models. A better understanding of these microscales' mechanisms can improve reservoir characterization and support the design of more effective water flooding strategies through improved predictive simulation at the pre-scale.

Future studies may explore the integration of artificial intelligence (AI) and real-time optimization to enhance the efficiency of water flooding under varying pore-scale and reservoir conditions

## Acknowledgment

We would like to acknowledge the School of Mining and Geosciences, Nazarbayev University, Astana, Kazakhstan, and China University of Petroleum, Changping, Beijing, China, for their support and permission to publish this significant research work.

## Nomenclature

|          |                                                                                           |
|----------|-------------------------------------------------------------------------------------------|
| $\delta$ | The Dirac delta function, used to restrict capillary forces to the fluid-fluid interface. |
| $\mu$    | Dynamic viscosity (Pa·s)                                                                  |
| $\nabla$ | Gradient operator                                                                         |
| $\rho$   | Density (kg/m <sup>3</sup> )                                                              |
| $F_c$    | Capillary force (N/m <sup>3</sup> )                                                       |
| $\gamma$ | Surface tension (N/m)                                                                     |
| $U$      | Velocity field (m/s)                                                                      |
| $U_r$    | Compression velocity (m/s)                                                                |
| $U_{cv}$ | Velocity in a particular control volume (m/s)                                             |
| $U_d$    | Maximum velocity in the porous medium (m/s)                                               |
| $\alpha$ | Compression coefficient (dimensionless)                                                   |
| $k$      | Interface curvature (1/m)                                                                 |
| $p$      | Pressure (Pa)                                                                             |
| $T_{Bk}$ | Brackbill number (dimensionless)                                                          |

## Funding

This research is funded by the School of Mining and Geosciences, Nazarbayev University, Astana, Kazakhstan.

## Conflict of Interest

The author hereby declares the absence of any conflict of interest

## References

- Ahmadi-Falavarjani, A., Mahani, H., & Ayatollahi, S. (2024). Pore-scale simulation of low-salinity waterflooding in mixed-wet systems: Effect of corner flow, surface heterogeneity, and kinetics of wettability alteration. *Scientific Reports*, **14**(1), 6563. <https://doi.org/10.1038/s41598-024-56846-0>
- Al-Raoush, R. I., & Willson, C. S. (2005). A pore-scale investigation of a multiphase porous media system. *Journal of Contaminant Hydrology*, **77**(1), 67–89. <https://doi.org/10.1016/j.jconhyd.2004.12.001>
- Chen, H., Sun, Z., Li, Y., & Su, H. (2025). Simulation of flow and heat transfer in high-temperature and high-pressure reservoir based on a multi-physical field coupling model at the pore scale. *The Canadian Journal of Chemical Engineering*, **103**(2), 914–926. <https://doi.org/10.1002/cjce.25389>
- Deshpande, S. S., Anumolu, L., & Trujillo, M. F. (2012). Evaluating the performance of the two-phase flow solver interFoam—IOPscience. <https://iopscience.iop.org/article/10.1088/1749-4699/5/1/014016>
- Elrahmani, A., Al-Raoush, R. I., Abugazia, H., & Seers, T. (2022). Pore-scale simulation of fine particles migration in porous media using coupled CFD-DEM. *Powder Technology*, **398**, 117130. <https://doi.org/10.1016/j.powtec.2022.117130>
- Fagbemi, S., Tahmasebi, P., & Piri, M. (2018). Interaction Between Fluid and Porous Media with Complex Geometries: A Direct Pore-Scale Study. <https://doi.org/10.1029/2017WR022242>
- Farhadzadeh, M., Bonto, M., & Nick, H. M. (2024). Pore-Scale Modeling of Heterogeneous Carbonate Rock Subjected to Modified Salinity Waterflooding. *Energy & Fuels*, **38**(8), 6821–6833. <https://doi.org/10.1021/acs.energyfuels.3c04103>
- Frank, F., Liu, C., Alpak, F. O., Berg, S., & Riviere, B. (2018). Direct Numerical Simulation of Flow on Pore-Scale Images Using the Phase-Field Method. *SPE Journal*, **23**(05), 1833–1850. <https://doi.org/10.2118/182607-PA>
- Hoang, D. A., van Steijn, V., Portela, L. M., Kreutzer, M. T., & Kleijn, C. R. (2013). Benchmark numerical simulations of segmented two-phase flows in microchannels using the Volume of Fluid method. *Computers & Fluids*, **86**, 28–36. <https://doi.org/10.1016/j.compfluid.2013.06.024>
- Ibrahim, M. M., & Bowden, S. A. (2021, December 9). Litho-Based Pore Scale Waterflooding in Carbonate Reservoirs: A Holistic Approach to Predict Displacement Efficiency. <https://dx.doi.org/10.2118/207443-MS>
- Jackson, M., Valvatne, P., & Blunt, M. (2003). Prediction of wettability variation and its impact on flow using pore-to reservoir-scale simulations | Semantic Scholar. <https://www.semanticscholar.org/paper/Prediction-of-wettability-variation-and-its-impact-Jackson-Valvatne/dbec0b6e1dc935cf716971fa0c91fd4242e8320d>
- Khan, M. J., Aziz, S., Asad, M., Kirmani, F. U. D., Aziz, H., & Alakbari, F. S. (2024). Investigating the Impact of Water and Polymer Flooding Techniques on Reservoir Performance in Heterogeneous Unconsolidated Formations: DOI: 10.14800/IOGR.1329. *Improved Oil and Gas Recovery*, **8**. <https://www.smartsitech.com/index.php/IOGR/article/view/1329>
- Khan, M. J., Mahmood, S. M., Alakbari, F. S., Siddiqui, N. A., Ridha, S., & Shafiq, M. U. (2024). Rock Wettability and Its Implication for Caprock Integrity in CO<sub>2</sub>-Brine Systems: A Comprehensive Review. *Energy & Fuels*, **38**(21), 19966–19991. <https://doi.org/10.1021/acs.energyfuels.4c02736>
- Khan, M. J., Muther, T., Aziz, H., & Mubeen-ur-Rehman, M. (2021). Investigating the impact of injection-water salinity and well strategies on water mobility and oil production in an oil-wet reservoir. *Modeling Earth Systems and Environment*, **7**(1), 247–260. <https://doi.org/10.1007/s40808-020-00934-3>
- Khan, M. J., Tunio, A. H., Ghirano, N. A., Muther, T., & Aziz, H. (2018, June 29). (PDF) An Overview of Low Saline Water Flooding to Improve Oil Recovery. ResearchGate. [https://www.researchgate.net/publication/333078841\\_An\\_Overview\\_of\\_Low\\_Saline\\_Water\\_Flooding\\_to\\_Improve\\_Oil\\_Recovery](https://www.researchgate.net/publication/333078841_An_Overview_of_Low_Saline_Water_Flooding_to_Improve_Oil_Recovery)
- Kyrloglou, A., Giefer, P., & Fritsching, U. (2024). *Frontiers | Numerical simulation of droplet dispersion within mesoporous membranes*. <https://doi.org/10.3389/fphy.2024.1282122>
- Li, Q., & Pistorius, P. C. (2023). Large Bubble-Resolved Direct Numerical Simulation for Multiphase Flow Applied to Gas-Stirred Ladles: Grid Resolution and Plug Eccentricity Effects. *Metallurgical and Materials Transactions B*, **54**(3), 1290–1313. <https://doi.org/10.1007/s11663-023-02762-z>
- Liu, G., Ding, Y., Wang, J., Ge, L., Chen, X., & Yang, D. (2023). Effect of pore-throat structure on air-foam flooding performance in a low-permeability reservoir. *Fuel*, **349**, 128620. <https://doi.org/10.1016/j.fuel.2023.128620>
- Liu, Y. F., & Jeng, D.-S. (2019). Pore scale study of the influence of particle geometry on soil permeability. *Advances in Water Resources*, **129**, 232–249. <https://doi.org/10.1016/j.advwatres.2019.05.024>
- Mahmud, W. M., Arns, J. Y., Sheppard, A., Knackstedt, M. A., & Pinczewski, W. V. (2007). Effect of network topology on two-phase imbibition relative permeability. *Transport in Porous Media*, **66**(3), 481–493. <https://doi.org/10.1007/s11242-006-0026-8>
- Mehmani, Y., & Tchelepi, H. A. (2017). Minimum requirements for predictive pore-network modeling of solute transport in micromodels. *Advances in Water Resources*, **108**, 83–98. <https://doi.org/10.1016/j.advwatres.2017.07.014>
- Mohammadi, M., & Mahani, H. (2020). Direct insights into the pore-scale mechanism of low-salinity waterflooding in carbonates using a novel calcite microfluidic chip. *Fuel*, **260**, 116374. <https://doi.org/10.1016/j.fuel.2019.116374>



- Mohammadmoradi, P., & Kantzas, A. (2016). *Pore Scale Investigation of Wettability Effect on Waterflood Performance*. Day 2 Tue, September 27, 2016, D021S036R007. <https://doi.org/10.2118/181309-MS>
- Molins, S. (2015). Reactive interfaces in direct numerical simulation of pore-scale processes. *Reviews in Mineralogy and Geochemistry*, **80**(1), 461–481.
- Namaee-Ghasemi, A., Ayatollahi, S., & Mahani, H. (2021). Pore-scale simulation of the interplay between wettability, capillary number, and salt dispersion on the efficiency of oil mobilization by low-salinity waterflooding. *SPE Journal*, **26**(06), 4000–4021.
- Pavuluri, S., Maes, J., & Doster, F. (2018). Spontaneous imbibition in a microchannel: Analytical solution and assessment of volume of fluid formulations. *Microfluidics and Nanofluidics*, **22**(8), Article 8. <https://doi.org/10.1007/s10404-018-2106-9>
- Qu, Y. G., & Liu, Y. T. (2012). The Impact of Sandstone's Geometry Form on Water Flooding Efficiency in Heterogeneous Reservoirs. *Petroleum Science and Technology*, **30**(17), 1813–1822. <https://doi.org/10.1080/10916466.2010.511390>
- Raeini, A. Q., Blunt, M. J., & Bijeljic, B. (2012). Modelling two-phase flow in porous media at the pore scale using the volume-of-fluid method. *Journal of Computational Physics*, **231**(17), 5653–5668. <https://doi.org/10.1016/j.jcp.2012.04.011>
- Selem, A. M., Agenet, N., Gao, Y., Raeini, A. Q., Blunt, M. J., & Bijeljic, B. (2021). Pore-scale imaging and analysis of low salinity waterflooding in a heterogeneous carbonate rock at reservoir conditions. *Scientific Reports*, **11**(1), 15063. <https://doi.org/10.1038/s41598-021-94103-w>
- Su, J., Chai, G., Wang, L., Cao, W., Gu, Z., Chen, C., & Xu, X. Y. (2019). Pore-scale direct numerical simulation of particle transport in porous media. *Chemical Engineering Science*, **199**, 613–627. <https://doi.org/10.1016/j.ces.2019.01.033>
- Su, J., Chai, G., Wang, L., Cao, W., Yu, J., Gu, Z., & Chen, C. (2019). Direct numerical simulation of pore scale particle-water-oil transport in porous media. *Journal of Petroleum Science and Engineering*, **180**, 159–175. <https://doi.org/10.1016/j.petrol.2019.04.078>
- Szűcs, H., Vehovszky, B., & Sanders, B. (2023). A Comprehensive Review of the Simulation Methods for Analysis at the Pore-scale. *Periodica Polytechnica Transportation Engineering*, **51**(2), Article 2. <https://doi.org/10.3311/PPtr.18452>
- Tian, Y., Ju, B., Chen, X., Chen, Z., Dong, Y., & Wu, D. (2021). Pore-Scale Investigation of Waterflooding Based on Experiments and Numerical Simulations Considering the Change in Geometry and Wettability. *Energy & Fuels*, **35**(21), 17617–17628. <https://doi.org/10.1021/acs.energyfuels.1c02682>
- van Leer, B. (1979). Towards the ultimate conservative difference scheme. V. A second-order sequel to Godunov's method. *Journal of Computational Physics*, **32**(1), 101–136. [https://doi.org/10.1016/0021-9991\(79\)90145-1](https://doi.org/10.1016/0021-9991(79)90145-1)
- Wang, Y., Song, R., Liu, J.-J., Cui, M.-M., & Ranjith, P. G. (2019). Pore scale investigation on scaling-up micro-macro capillary number and wettability on trapping and mobilization of residual fluid. *Journal of Contaminant Hydrology*, **225**, 103499. <https://doi.org/10.1016/j.jconhyd.2019.103499>
- Wood, B. D., Apte, S. V., Liburdy, J. A., Ziazi, R. M., He, X., Finn, J. R., & Patil, V. A. (2015). A comparison of measured and modeled velocity fields for a laminar flow in a porous medium. *Advances in Water Resources*, **85**, 45–63. <https://doi.org/10.1016/j.advwatres.2015.08.013>
- Yang, X., Mehmani, Y., Perkins, W. A., Pasquali, A., Schönherr, M., Kim, K., Perego, M., Parks, M. L., Trask, N., Balhoff, M. T., Richmond, M. C., Geier, M., Krafczyk, M., Luo, L.-S., Tartakovsky, A. M., & Scheibe, T. D. (2016). Intercomparison of 3D pore-scale flow and solute transport simulation methods. *Advances in Water Resources*, **95**, 176–189. <https://doi.org/10.1016/j.advwatres.2015.09.015>
- Zhang, J., Liu, X., Chen, D., & Yin, Z. (2022). An investigation on the permeability of hydrate-bearing sediments based on pore-scale CFD simulation. *International Journal of Heat and Mass Transfer*, **192**, 122901. <https://doi.org/10.1016/j.ijheatmasstransfer.2022.122901>